CAPE DC/AC Control Design

Grid-connected inverter with LCL filter

Shahan
Master in Electronic Engineering — UPC Barcelona
Course academic staff: Domingo Biel, Francesc Guinjoan

Deadline: June 12th, 2026

Filter parameters: $L_1 = 2\,\text{mH}$, $C = 10\,\mu\text{F}$, $L_2 = 1\,\text{mH}$.
Controlled variable: grid-side current i_2. **Grid frequency:** $f_g = 50\,\text{Hz}$ (nominal), varying in $[48\,\text{Hz}, 52\,\text{Hz}]$.

1 Q1 – Uncompensated open-loop transfer function (2 p)

For a lossless LCL filter, applying KVL/KCL with the inverter voltage V_i as input and the grid current I_2 as output (with $V_g = 0$ for the control TF) yields:

$$\boxed{G(s) = \frac{I_2(s)}{V_i(s)} = \frac{1}{s^3 L_1 L_2 C + s\,(L_1 + L_2)}} \tag{1}$$

Resonant frequency. The two complex-conjugate poles of (1) lie at $s = \pm j\omega_{res}$, where

$$\omega_{res} = \sqrt{\frac{L_1 + L_2}{L_1 L_2 C}} = \sqrt{\frac{3 \cdot 10^{-3}}{2 \cdot 10^{-11}}} = 12\,247.45\,\text{rad}\,\text{s}^{-1}, \qquad f_{res} = \frac{\omega_{res}}{2\pi} = \boxed{1949.24\,\text{Hz}}. \tag{2}$$

Resonant peak. For an *ideal* (lossless) LCL filter, the closed-form magnitude at $\omega = \omega_{res}$ is unbounded ($|G(j\omega_{res})| \to \infty$) because the poles lie exactly on the imaginary axis. In MATLAB, the value returned by `bode` or `getPeakGain` depends on how close the frequency grid samples the exact resonance: with the default grid, the reported peak is typically of the order of +20 to +30 dB, but it grows without bound as the grid is refined. The dashed line in Fig. 1 shows the sampled-grid peak from MATLAB ($\approx +22.8$ **dB**), while the analytical result is **theoretically infinite**. This is precisely the reason a passive (or active) damping technique is required, which is addressed in question Q2.

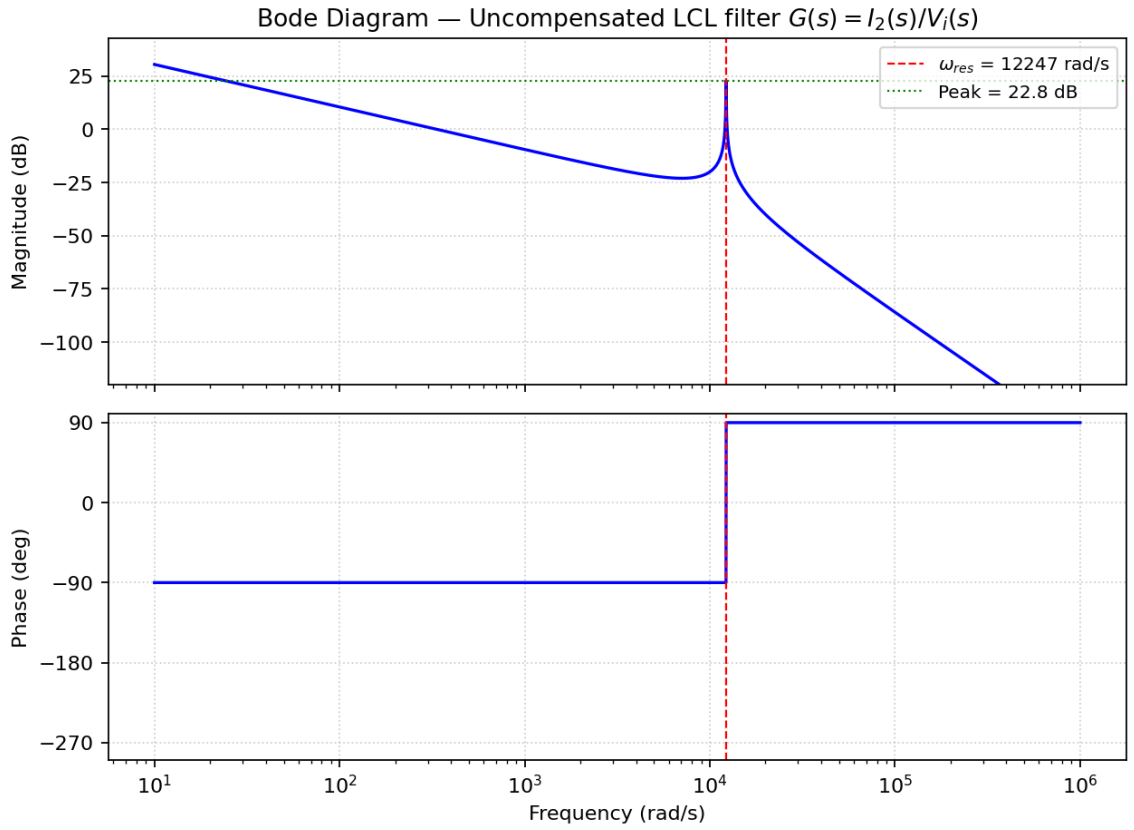


Figure 1: Bode diagram of the uncompensated LCL filter $G(s)$. The phase jumps by 180° at the resonance (purely imaginary poles), and the magnitude exhibits an undamped peak whose theoretical value is $+\infty$.

2 Q2 – Passive damping with R_C in series with C (2 p)

Placing a resistor R_C in series with the filter capacitor modifies the transfer function by adding a zero in the numerator and a quadratic term in the denominator:

$$G_d(s) = \frac{I_2(s)}{V_i(s)} = \frac{1 + s\,R_C C}{s^3 L_1 L_2 C \;+\; s^2 R_C C (L_1 + L_2) \;+\; s\,(L_1 + L_2)} \tag{3}$$

The new s^2 term gives the poles a real (negative) part, which limits the magnitude at the resonance to a finite value.

Numerical sweep. Sweeping R_C in MATLAB and looking for the maximum of $|G_d(j\omega)|$ in the band $0.3\,\omega_{res} \le \omega \le 3\,\omega_{res}$ produces the curve in Fig. 2. The smallest R_C for which the resonant peak just touches -20 dB is

$$\boxed{R_C \;\approx\; 2.42\,\Omega} \tag{4}$$

For larger values of R_C the magnitude curve has no local maximum above -20 dB at all (it simply blends with the high-frequency asymptote), and for smaller values the peak grows back above -20 dB.

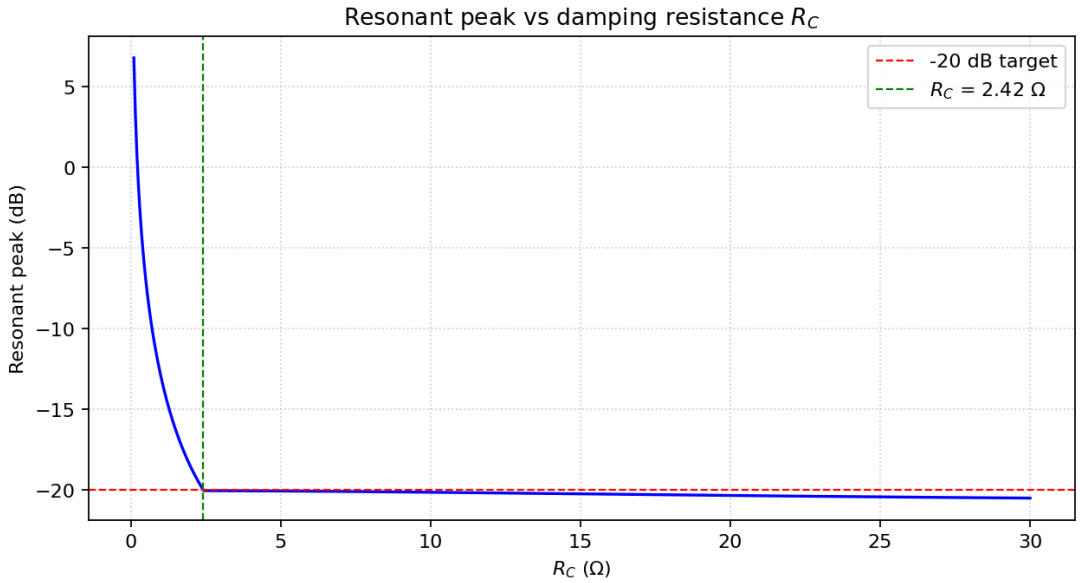


Figure 2: Resonant peak of $G_d(s)$ as a function of R_C (MATLAB sweep). The horizontal red dashed line marks the $-20\,\mathrm{dB}$ target.

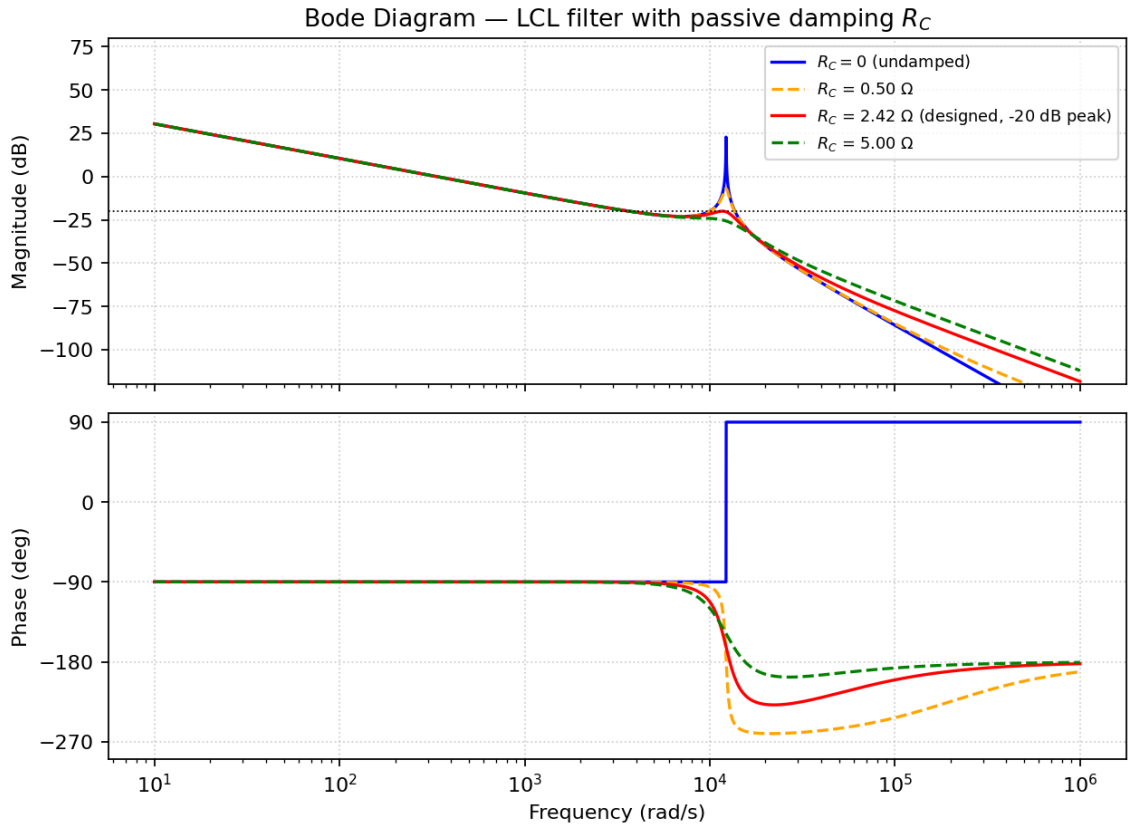


Figure 3: Bode diagram of the LCL filter with several values of R_C. With the chosen $R_C \approx 2.42\,\Omega$ (red, solid) the peak is flattened to $-20\,\mathrm{dB}$ and the phase transition becomes smooth, recovering the $-20\,\mathrm{dB/dec}$ slope and $\approx -180°$ phase at high frequency.

3 Q3 – Why a Proportional + Resonant controller? (2 p)

The current reference for a grid-connected inverter is sinusoidal at the grid frequency ω_g: $i_{\text{ref}}(t) = \hat{I}\sin(\omega_g t + \varphi)$. For the closed loop to have *zero steady-state error* on such an AC reference (and to perfectly reject sinusoidal disturbances of the same frequency, e.g. grid harmonics or grid voltage), the **Internal Model Principle** requires the controller to contain the model of the reference inside its open-loop transfer function:

$$G_{\text{ref}}(s) = \frac{\omega_g}{s^2 + \omega_g^2} \quad \Longrightarrow \quad G_C(s) \text{ must have poles at } s = \pm j\omega_g. \tag{5}$$

The proposed PR controller exactly satisfies this:

$$G_C(s) \;=\; k_p \;+\; \frac{k_i\, s}{s^2 + \omega_g^2} \;=\; k_p\, \frac{s^2 + (k_i/k_p)\, s + \omega_g^2}{s^2 + \omega_g^2}, \tag{6}$$

because its denominator $s^2+\omega_g^2$ places two undamped poles at $\pm j\omega_g$. As a consequence $|G_C(j\omega_g)| \to \infty$, which is the AC analogue of the integral action of a PI controller (infinite gain at DC).

Additional advantages over a PI in dq:

- Works directly in the stationary $\alpha\beta$ frame: no Park transformation, no PLL synchronisation in the control path, no d/q cross-coupling compensation.
- Single-phase systems and unbalanced three-phase systems are handled identically (no need for a synchronous reference frame, which fails for negative-sequence components).
- Lower computational cost and simpler digital implementation than two coupled PI controllers + Park transform.
- Multiple resonant terms can be stacked at ω_g, $3\omega_g$, $5\omega_g$, $7\omega_g$, … to selectively reject low-order grid harmonics with the same architecture.

For all these reasons, the PR controller is the standard choice for grid-connected inverters with sinusoidal current control.

4 Q4 – Design of k_p (2 p)

Design approach. The PR controller behaves as a pure gain k_p at frequencies sufficiently above ω_g (the resonant term decays as $1/\omega$). The damped LCL plant, in the same frequency band and below ω_{res}, behaves asymptotically as $G_d(s) \;\approx\; \dfrac{1}{s(L_1+L_2)}$, i.e. a pure integrator with a $-20\,\text{dB/dec}$ slope. Hence the loop-gain magnitude is approximately

$$|L(j\omega)| \;=\; |G_C(j\omega)|\,|G_d(j\omega)| \;\approx\; \frac{k_p}{\omega(L_1+L_2)}, \qquad \omega_g \ll \omega \ll \omega_{res}. \tag{7}$$

Imposing $|L(j\omega_c)| = 1$ at a chosen crossover frequency ω_c gives the design rule

$$\boxed{k_p \;=\; \omega_c\,(L_1+L_2)} \tag{8}$$

Choice of ω_c. To guarantee a robust and well-damped closed loop, ω_c must sit:

- Well *above* ω_g, so that the PR controller acts as a pure k_p at the crossover (otherwise the resonant peak of G_C would distort the phase margin).
- Well *below* ω_{res}, to keep the loop in the clean $-20\,\text{dB/dec}$ region and away from the LCL resonance, where the phase rapidly drops towards $-180°$.

A widely used choice is $\omega_c \approx \omega_{res}/10$:

$$\omega_c = \frac{\omega_{res}}{10} = 1224.7\,\mathrm{rad\,s^{-1}} \quad (f_c \approx 195\,\mathrm{Hz}), \tag{9}$$

$$k_p = \omega_c\,(L_1 + L_2) = 1224.7 \cdot 3 \times 10^{-3}\,\mathrm{H} \;=\; \boxed{k_p \approx 3.67} \tag{10}$$

This places the crossover comfortably between the grid frequency (50 Hz) and the resonance (1949 Hz), in a ratio of approximately 4 : 1 on each side. The MATLAB function `margin(kp*Gd)` verifies that the loop gain crosses 0 dB at $f_c \approx 197$ Hz with a phase margin of $\approx 90°$ (Fig. 4), which is an excellent stability margin.

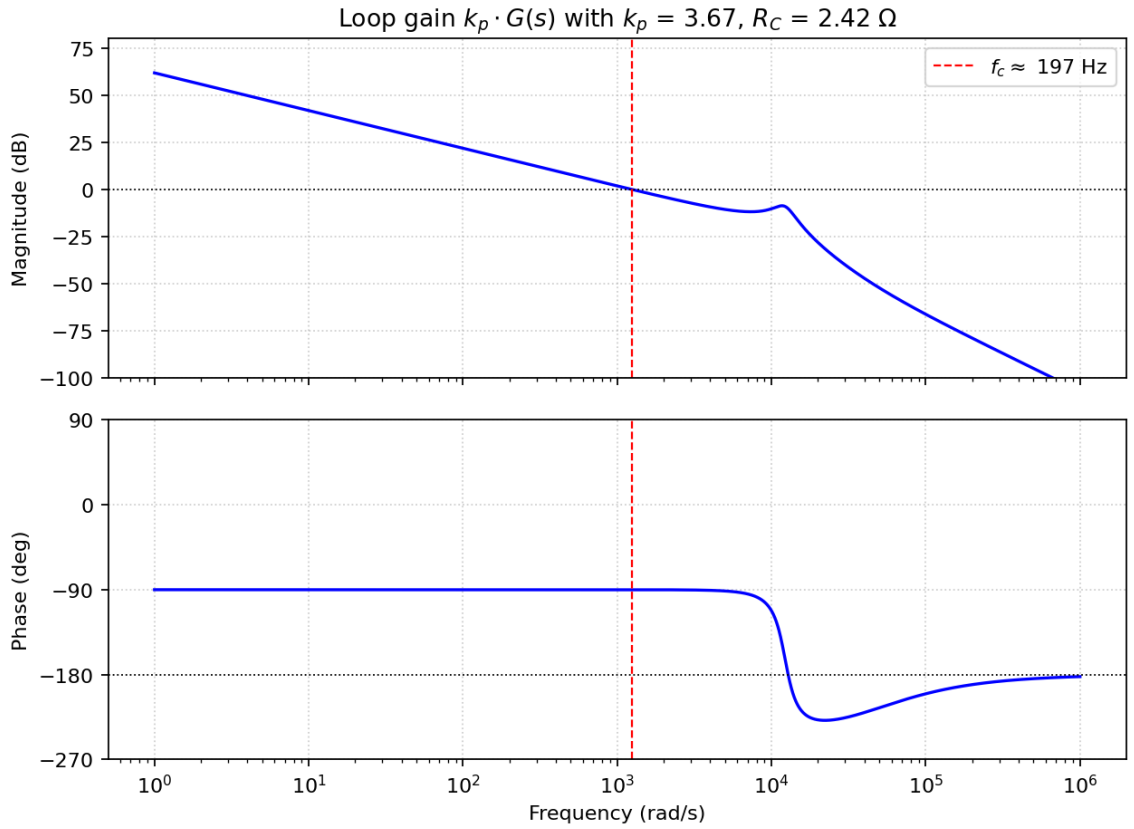


Figure 4: Loop gain $L(s) = k_p\,G_d(s)$ with $k_p = 3.67$ and $R_C = 2.42\,\Omega$. Crossover at ≈ 197 Hz, phase margin $\approx 90°$.

5 Q5 – Valid values of k_i for the 50 dB requirement (2 p)

The figure provided in the assignment plots $|G_C(j2\pi f)|$ for $k_p = 1$ and $k_i \in \{50, 250, 1000, 5000\}$ in the band $45 \le f \le 55$ Hz. With $k_p = 1$:

$$|G_C(j\omega)| \;=\; \sqrt{1 + \left(\frac{k_i\,\omega}{\omega_g^2 - \omega^2}\right)^2}. \tag{11}$$

The worst case in the interval $[48, 52]$ Hz occurs at the edges ($f = 48$ or $f = 52$ Hz), where $|\omega_g^2 - \omega^2|$ is largest and hence the controller gain is smallest. Evaluating (11) analytically:

Table 1: Controller magnitude $|G_C|$ at the edges of the 48–52 Hz band, with $k_p = 1$.

k_i	$\lvert G_C(j2\pi\cdot 48)\rvert$ [dB]	$\lvert G_C(j2\pi\cdot 52)\rvert$ [dB]	≥ 50 dB?
50	6.81	7.09	No
250	19.82	20.16	No
1000	31.82	32.17	No
5000	**45.80**	**46.14**	Close (–4 dB)

Reading the Bode plot in Fig. 5. Only the $k_i = 5000$ curve (red) approaches the 50 dB horizontal threshold inside the orange band $[48, 52]$ Hz; in fact it crosses the 50 dB line at ≈ 48.75 and ≈ 51.27 Hz. Therefore:

- $k_i = 50, 250, 1000$ **do not** satisfy the requirement (peak gains well below 50 dB at the edges of the band).
- $k_i = 5000$ is the **only candidate among the four**: it provides ≥ 50 dB for f inside $\approx [48.75, 51.27]$ Hz. However it is still ≈ 4 dB short at the very edges $f = 48$ and $f = 52$ Hz.

Conclusion. Among the four proposed values, $\boxed{k_i = 5000}$ is the *best* choice (and the only one that comes close to the 50 dB specification). To strictly meet $|G_C| \geq 50$ dB over the entire $[48, 52]$ Hz band one would need $k_i \geq 8120$, which is obtained from (11) by imposing $\left|k_i 2\pi 48 / (\omega_g^2 - (2\pi 48)^2)\right| = 10^{50/20}$.

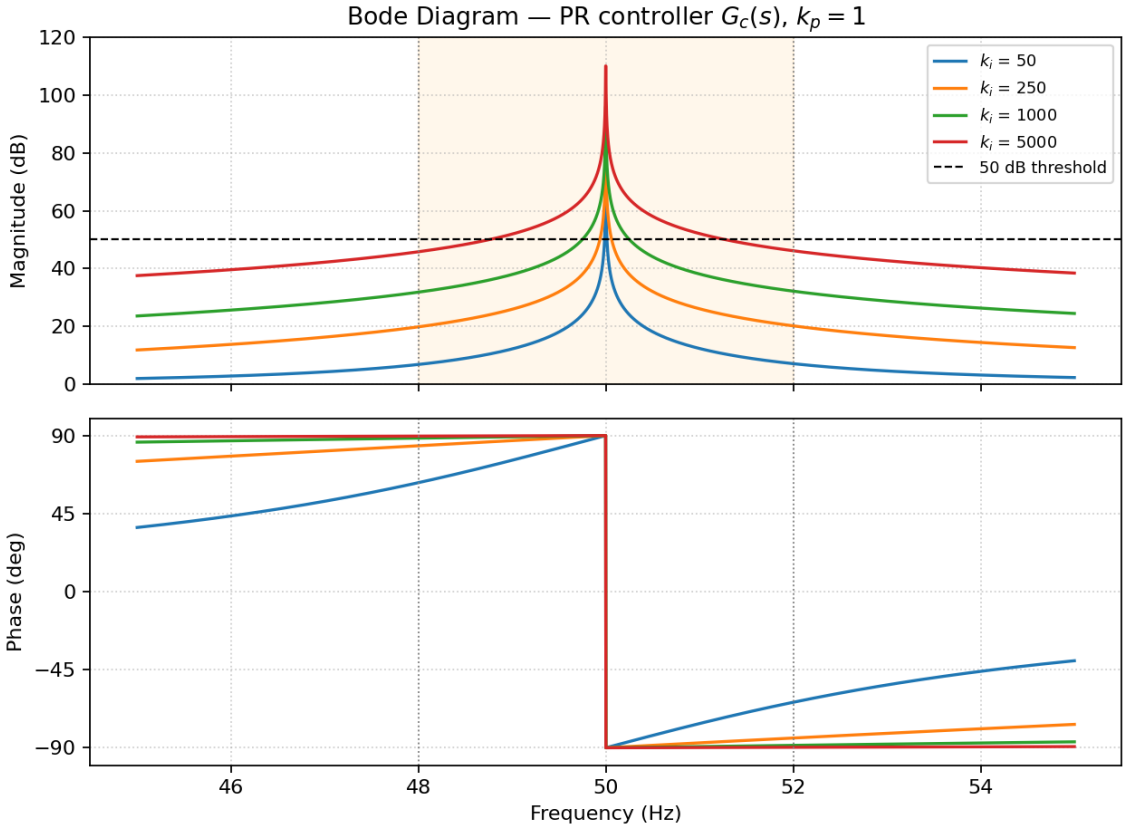


Figure 5: Bode diagram of the PR controller $G_C(s)$ with $k_p = 1$ and $k_i \in \{50, 250, 1000, 5000\}$ (reproduction of the figure of the assignment). The dashed black line is the 50 dB threshold; the orange band highlights the $[48, 52]$ Hz grid-frequency range. Only $k_i = 5000$ approaches the threshold inside this band.

Summary of design values

Quantity	Value
Resonant frequency f_{res}	1949.24 Hz
Resonant pulsation ω_{res}	12 247.45 rad s^{-1}
Damping resistance R_C	2.42 Ω
Proportional gain k_p	3.67
Best k_i from $\{50, 250, 1000, 5000\}$	5000 (close to spec; ≥ 8120 for full compliance)
Crossover frequency f_c (designed)	$\approx$ 197 Hz
Phase margin	$\approx 90°$

A MATLAB script used for the simulations

The script below produces every figure of this delivery and prints the numerical results in the command window.

Listing 1: `CAPE_DC_AC.m`

```
%% CAPE DC/AC Control Design - Grid-connected inverter with LCL filter
%  Master in Electronic Engineering - UPC Barcelona
%  Author: Shahan (and Ayush)
%  ------------------------------------------------------------------
%  This script answers questions 1 to 5 of the delivery.
%  Run section by section (Ctrl+Enter) or all at once (F5).
%  ------------------------------------------------------------------

clear; clc; close all;

%% Filter parameters
L1 = 2e-3;      % H, inverter-side inductor
C  = 10e-6;     % F, filter capacitor
L2 = 1e-3;      % H, grid-side inductor

s = tf('s');

%% ================================================================
%  Q1 - Uncompensated open-loop transfer function and Bode plot
%  ================================================================
%
%  For a non-losses LCL filter, the TF from the inverter voltage Vi
%  to the grid-side current I2 (the controlled variable) is:
%
%              I2(s)              1
%      G(s) = ----- = ----------------------------
%              Vi(s)    s^3*L1*L2*C + s*(L1+L2)
%
G = 1 / ( s^3*L1*L2*C + s*(L1+L2) );

% Resonant frequency (analytical)
wres = sqrt( (L1+L2) / (L1*L2*C) );    % rad/s
fres = wres/(2*pi);                    % Hz
fprintf('Q1) wres = %.2f rad/s  ->  fres = %.2f Hz\n', wres, fres);

% Bode plot
figure('Name','Q1: Uncompensated LCL');
bode(G, {1e1, 1e6}); grid on;
title('Bode Diagram - Uncompensated LCL filter G(s) = I_2(s)/V_i(s)');

% Peak (MATLAB returns a sampling-grid value; the ideal lossless system
% has an infinite peak at exactly w = wres)
[gpeak, ~] = getPeakGain(G);
```

```
fprintf('   Resonant peak (sampled): %.2f dB (theoretically +inf)\n\n', 20*log10(gpeak));

%% ======================================================================
%  Q2 - Passive damping: find Rc such that the resonant peak is -20 dB
%  ======================================================================
%
%  Adding Rc in series with C gives:
%
%              I2(s)         1 + s*Rc*C
%   G_d(s) = ----- = ------------------------------------------
%              Vi(s)    s^3*L1*L2*C + s^2*Rc*C*(L1+L2) + s*(L1+L2)
%

% Sweep Rc and look for the resonant peak around wres
Rc_vec = linspace(0.1, 10, 1000);
peak_dB = zeros(size(Rc_vec));
w_eval  = logspace(2, 6, 4000);

for k = 1:length(Rc_vec)
    Rc = Rc_vec(k);
    Gd = (1 + s*Rc*C) / ( s^3*L1*L2*C + s^2*Rc*C*(L1+L2) + s*(L1+L2) );
    [mag,~] = bode(Gd, w_eval);
    mag = squeeze(mag);
    % Look only in the resonance band
    mask = (w_eval > 0.3*wres) & (w_eval < 3*wres);
    peak_dB(k) = 20*log10(max(mag(mask)));
end

% Find Rc that gives ~ -20 dB peak (smallest Rc that satisfies it)
[~, idx] = min(abs(peak_dB - (-20)));
Rc_opt = Rc_vec(idx);
fprintf('Q2) Rc that yields a -20 dB peak: Rc = %.2f ohm\n\n', Rc_opt);

% Plot peak vs Rc
figure('Name','Q2: Peak vs Rc');
plot(Rc_vec, peak_dB, 'b','LineWidth',1.4); hold on; grid on;
yline(-20,'r--','LineWidth',1.2,'Label','-20 dB target');
xline(Rc_opt,'g--','LineWidth',1.2,'Label',sprintf('R_C = %.2f \\Omega', Rc_opt));
xlabel('R_C (\Omega)'); ylabel('Resonant peak (dB)');
title('Resonant peak vs damping resistance R_C');

% Compare Bode: undamped vs damped
Gd_opt = (1 + s*Rc_opt*C) / ( s^3*L1*L2*C + s^2*Rc_opt*C*(L1+L2) + s*(L1+L2) );

figure('Name','Q2: Damped Bode');
bode(G, Gd_opt, {1e1, 1e6}); grid on;
legend('R_C = 0 (undamped)', ...
       sprintf('R_C = %.2f \\Omega (designed, -20 dB peak)', Rc_opt), ...
       'Location','SouthWest');
title('Bode Diagram - LCL filter with passive damping');

%% ======================================================================
%  Q3 - Justification: why use a Proportional + Resonant controller
%  ======================================================================
%
%  The reference current i_ref(t) is sinusoidal at the grid frequency
%  wg = 2*pi*50 rad/s. A standard PI controller (in the stationary
%  alpha-beta frame) cannot give zero steady-state error for AC
%  references because its infinite gain is located at s = 0 only.
%
%  The PR controller:
%       Gc(s) = kp + ki*s/(s^2 + wg^2)
%
%  places an infinite (unbounded) gain at s = +-j*wg, i.e. exactly
```

```
%  at the grid frequency. By the Internal Model Principle, this
%  guarantees zero steady-state error for sinusoidal references and
%  rejects sinusoidal disturbances of the same frequency.
%
%  Additional advantages versus PI in dq:
%   - No Park/Clarke transformation required (works in alpha-beta).
%   - No coupling between d/q axes.
%   - Lower computational cost and simpler implementation.
%
%  (See written justification in the PDF.)

%% ================================================================
%  Q4 - Design of kp
%  ================================================================
%
%  Above the grid frequency and well below the resonance, the PR
%  controller behaves as Gc(s) ~ kp, and the damped LCL plant behaves
%  asymptotically as G_d(s) ~ 1/(s*(L1+L2)).  The loop gain is then:
%
%       |L(jw)| = kp / ( w*(L1+L2) )
%
%  Forcing |L(jw_c)| = 1 at a desired crossover frequency w_c gives:
%
%       kp = w_c * (L1 + L2)
%
%  We choose w_c well below the resonance (a typical choice is
%  w_c = wres/10) so the closed loop is robust against the resonant
%  dynamics and has comfortable phase margin.

wc_target = wres/10;                % desired crossover (rad/s)
fc_target = wc_target/(2*pi);       % Hz
kp = wc_target * (L1 + L2);
fprintf('Q4) Designed kp = %.3f  (target fc = %.1f Hz, target wc = %.1f rad/s)\n', ...
        kp, fc_target, wc_target);

% Loop gain with the designed kp (PR resonant part has no effect here
% because the grid frequency is far from the crossover region)
L_loop = kp * Gd_opt;

figure('Name','Q4: Loop gain');
margin(L_loop); grid on;
title(sprintf('Loop gain k_p*G_d(s) with k_p = %.2f, R_C = %.2f \\Omega', kp, Rc_opt));

%% ================================================================
%  Q5 - PR controller: which ki satisfies |Gc| >= 50 dB on 48-52 Hz?
%  ================================================================
%
%  Gc(s) = kp + ki*s/(s^2 + wg^2),   kp = 1
%
%  At s = j*w:
%   |Gc(jw)| = sqrt( kp^2 + ( ki*w / (wg^2 - w^2) )^2 )

wg = 2*pi*50;        % grid frequency in rad/s (nominal)
ki_list = [50, 250, 1000, 5000];

% Reproduce the figure (kp = 1)
figure('Name','Q5: PR controller');
hold on; grid on;
f_axis = linspace(45, 55, 4000);
w_axis = 2*pi*f_axis;
colors = lines(4);
for i = 1:length(ki_list)
    ki = ki_list(i);
    Gc = 1 + ki*s/(s^2 + wg^2);
```

```
    [magc,~] = bode(Gc, w_axis);
    magc = squeeze(magc);
    plot(f_axis, 20*log10(magc), 'Color', colors(i,:), 'LineWidth', 1.5, ...
        'DisplayName', sprintf('k_i = %d', ki));
end
yline(50,'k--','LineWidth',1.2,'DisplayName','50 dB threshold');
xline(48,':','Color',[0.5 0.5 0.5]); xline(52,':','Color',[0.5 0.5 0.5]);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
title('Bode Diagram - PR controller (k_p = 1)');
legend('Location','NorthEast');
ylim([0 120]);

% Check the worst-case gain (at f = 48 or 52 Hz)
fprintf('\nQ5) Worst-case gains at the edges of [48,52] Hz (kp = 1):\n');
fprintf('    ki   |  |Gc(48Hz)|  |  |Gc(52Hz)|  |  meets >=50 dB?\n');
fprintf('   ------+--------------+---------------+------------------\n');
for ki = ki_list
    g48 = sqrt(1 + (ki*2*pi*48/(wg^2 - (2*pi*48)^2))^2);
    g52 = sqrt(1 + (ki*2*pi*52/(wg^2 - (2*pi*52)^2))^2);
    g48_dB = 20*log10(g48);
    g52_dB = 20*log10(g52);
    meets = "NO ";
    if min(g48_dB, g52_dB) >= 50, meets = "YES"; end
    fprintf('  %5d  |   %6.2f dB  |    %6.2f dB |  %s\n', ki, g48_dB, g52_dB, meets);
end

%% ========================================================================
%  Summary
%  ========================================================================
fprintf('\n=================== FINAL DESIGN VALUES ===================\n');
fprintf(' Resonant frequency :  wres = %.0f rad/s  =  fres = %.2f Hz\n', wres, fres);
fprintf(' Damping resistor   :  Rc   = %.2f ohm\n', Rc_opt);
fprintf(' Proportional gain  :  kp   = %.2f   (fc ~ fres/10 = %.0f Hz)\n', kp, fc_target);
fprintf(' Best ki of {50,250,1000,5000} for the >=50 dB spec: 5000\n');
fprintf('============================================================\n');
```